

Nima Ghourchian

Karaj, Iran

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Education

Islamic Azad University – Karaj Branch

Sep 2021 – Feb 2026

Bachelor of Mechanical Engineering | GPA: 16.90/20

Relevant coursework: Statics: 18/20, Dynamics: 17/20, Mechanical vibrations: 19.5/20, Automatic Control: 19/20, Machine Dynamics: 20/20, Engineering Mathematics: 18/20, Differential Equations: 20/20, Numerical Analysis: 19/20, Fundamentals of Electrical Engineering 1: 20/20

Selected Projects

National CubeSat Design Competition – Qsat

Jan 2024 – Jul 2026

Attitude Determination & Control Team member | Sharif University Team (Cubisa)

- Completed a 180-hour CubeSat Design workshop covering topics such as ADCS, orbital mechanics, and mission design and analysis.
- Researched and implemented B-dot detumbling and quaternion-based PD feedback controller for nadir pointing in Simulink.
- Conducted Software-in-the-Loop (SIL) and Processor-in-the-Loop (PIL) testing to verify control algorithms and embedded implementation.

Vision-Based GPS-denied Navigation for Emergency Response UAVs

Jun 2025 – Jan 2026

B.Sc. Final Project

- Built a localization pipeline to match simulated UAV imagery with georeferenced satellite images using ResNet and LiteSAM to generate latitude and longitude estimates.
- Designed and implemented a Kalman filter that fuses simulated IMU, barometer and vision-based pose measurements.

Professional Experience

SpaceOmid

Sep 2025 – Apr 2026

Attitude Determination & Control Engineer

- Contributed to the development of a star tracker for spacecraft attitude determination.
- Developed a camera-based module to measure the ground-truth orientation of a satellite testbed using AprilTags.

Flight Control & Simulation Engineer – Omidspac Team

- Collaborated with a team of engineers to test and debug a spraying module prototype.
- Extended the existing manual altitude-control system by integrating an obstacle-responsive module that enabled automatic altitude adjustment during flight.

SpaceOmid

Jul 2025 – Sep 2025

Intern – Omidspac Team

- Participated in and contributed to weekly technical meetings.
- Assisted in sensor calibration, PID tuning of flight controllers and regular test flights.

Teaching Experience

Teaching Assistant | Islamic Azad University | Supervised by Dr. Mirzadeh

Fall 2025

Automatic Control

- conducted teaching sessions, led class discussions, and provided support in understanding complex concepts.

Volunteer Mathematics Instructor | Karaj

Sep 2024 – Jun 2025

Algebra and Geometry

- Taught foundational algebra and geometry to middle-school students through a volunteer educational program.

Honors and Awards

- **Ranked 73rd** among more than 10,000 participants in the Iranian National Graduate Entrance Exam for Mechanical Engineering.
- **Placed 3rd** among more than 50 teams in Qsat National CubeSat Design Competition.

Technical Skills

Programming Languages: MATLAB/Simulink, Python, C/C++

Frameworks & Tools: ROS 2, ArduPilot, PX4, AirSim, Unreal Engine

Software: SolidWorks, Proteus, STK

Languages

Persian: Native

English: IELTS Academic **7.0** (C1 Level)

Other Projects

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|-----------------------------------|---|
| • H-Bridge DC Motor Driver Design | • Temperature Control system Design |
| • Facial Detection Door Lock | • Peaucellier Mechanism Design |
| • Reduction Gearbox CAD Design | • Target Tracking and Capture in ROS2 Turtlesim |